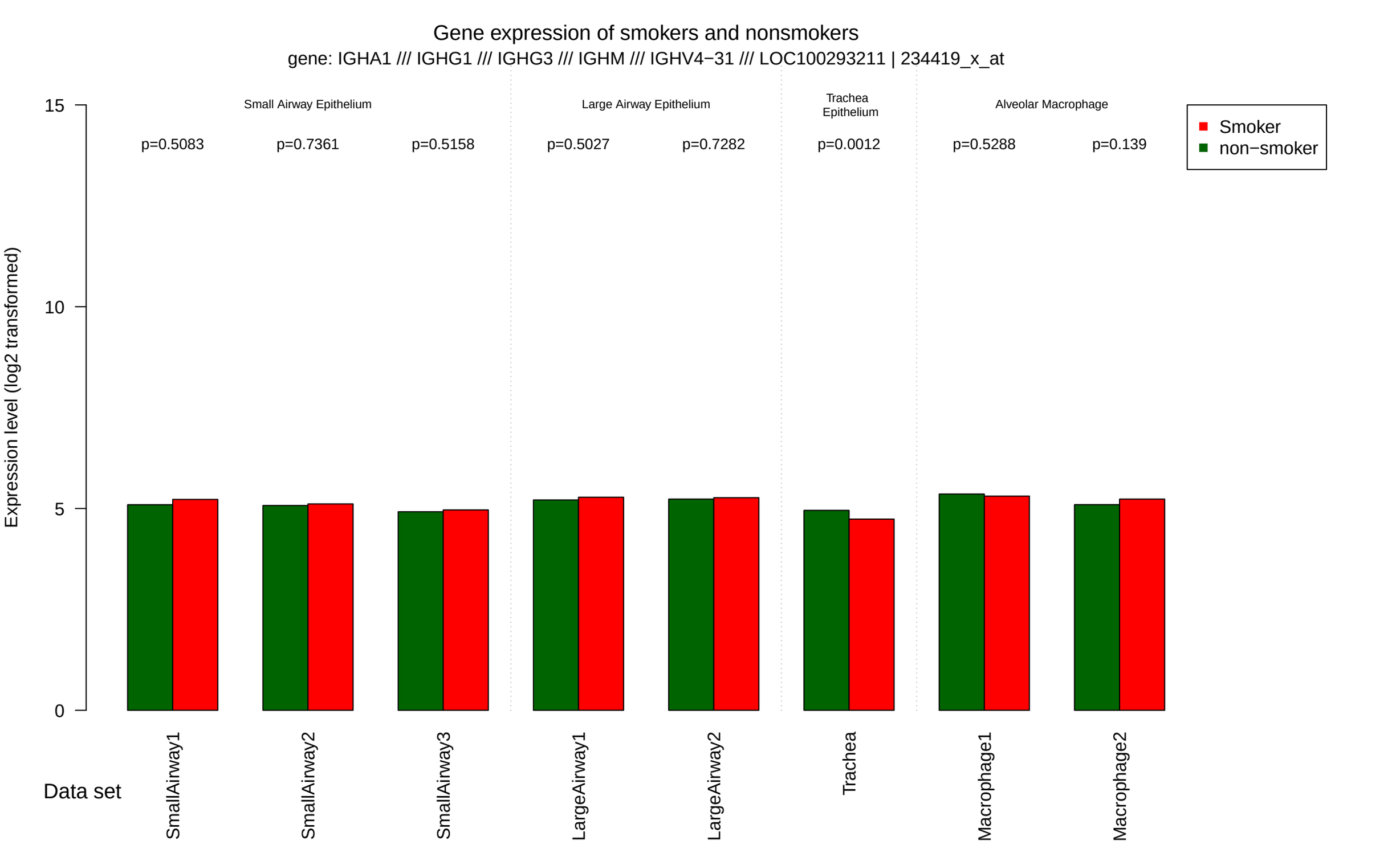
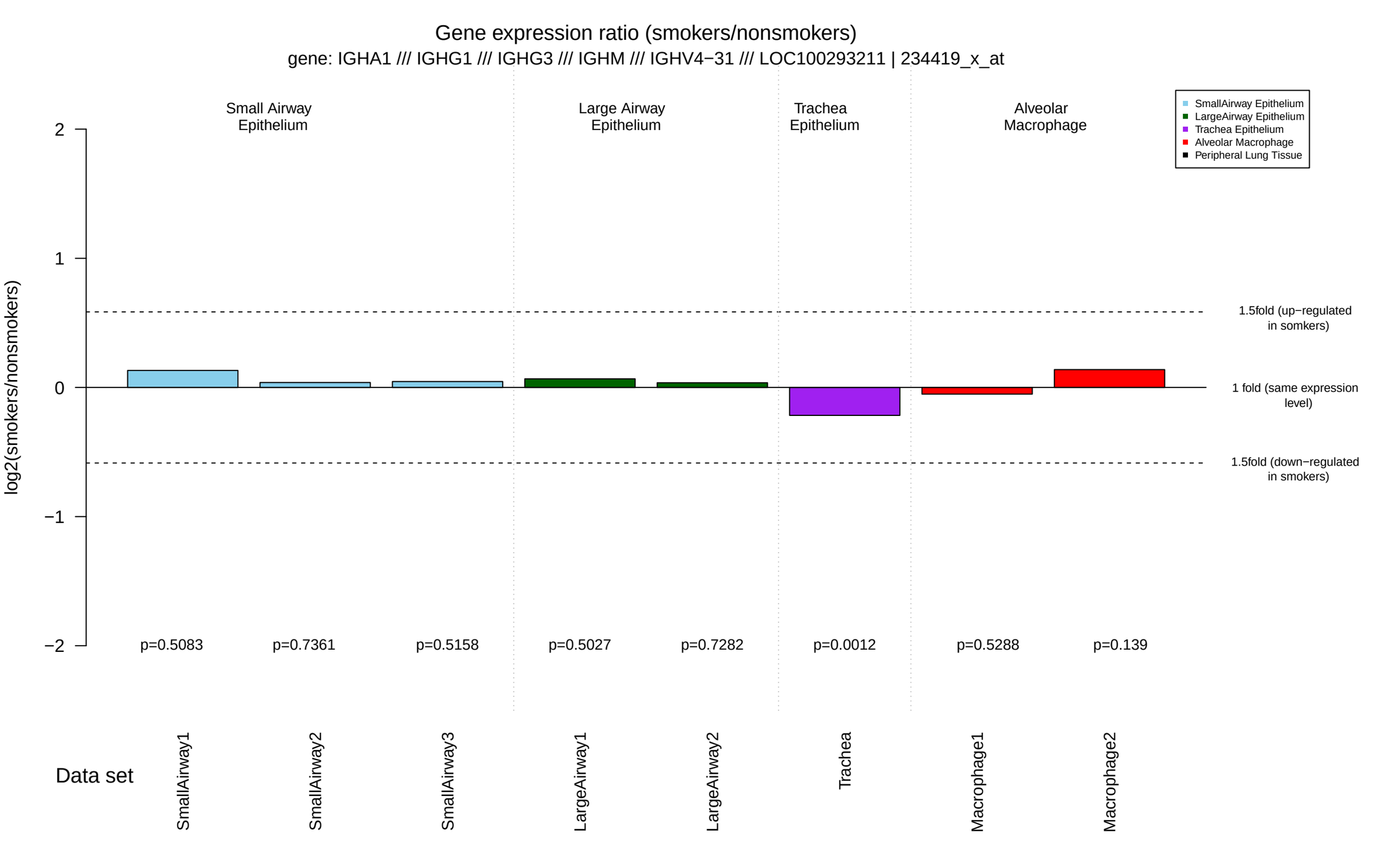
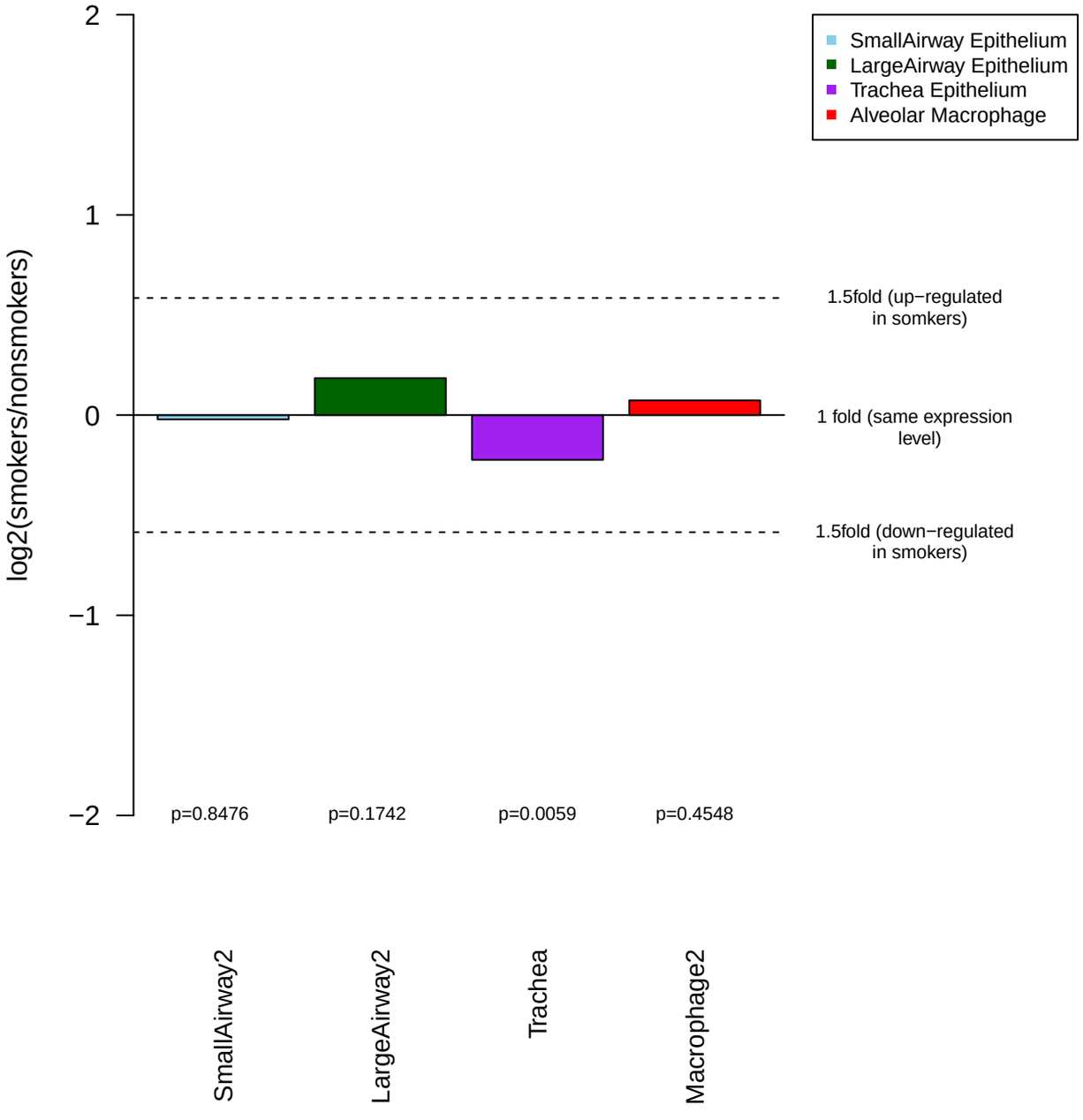

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5083
p=0.7361
p=0.5158
p=0.5027
p=0.7282
p=0.0012
p=0.5288
p=0.139
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5083
p=0.7361
p=0.5158
p=0.5027
p=0.7282
p=0.0012
p=0.5288
p=0.139
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

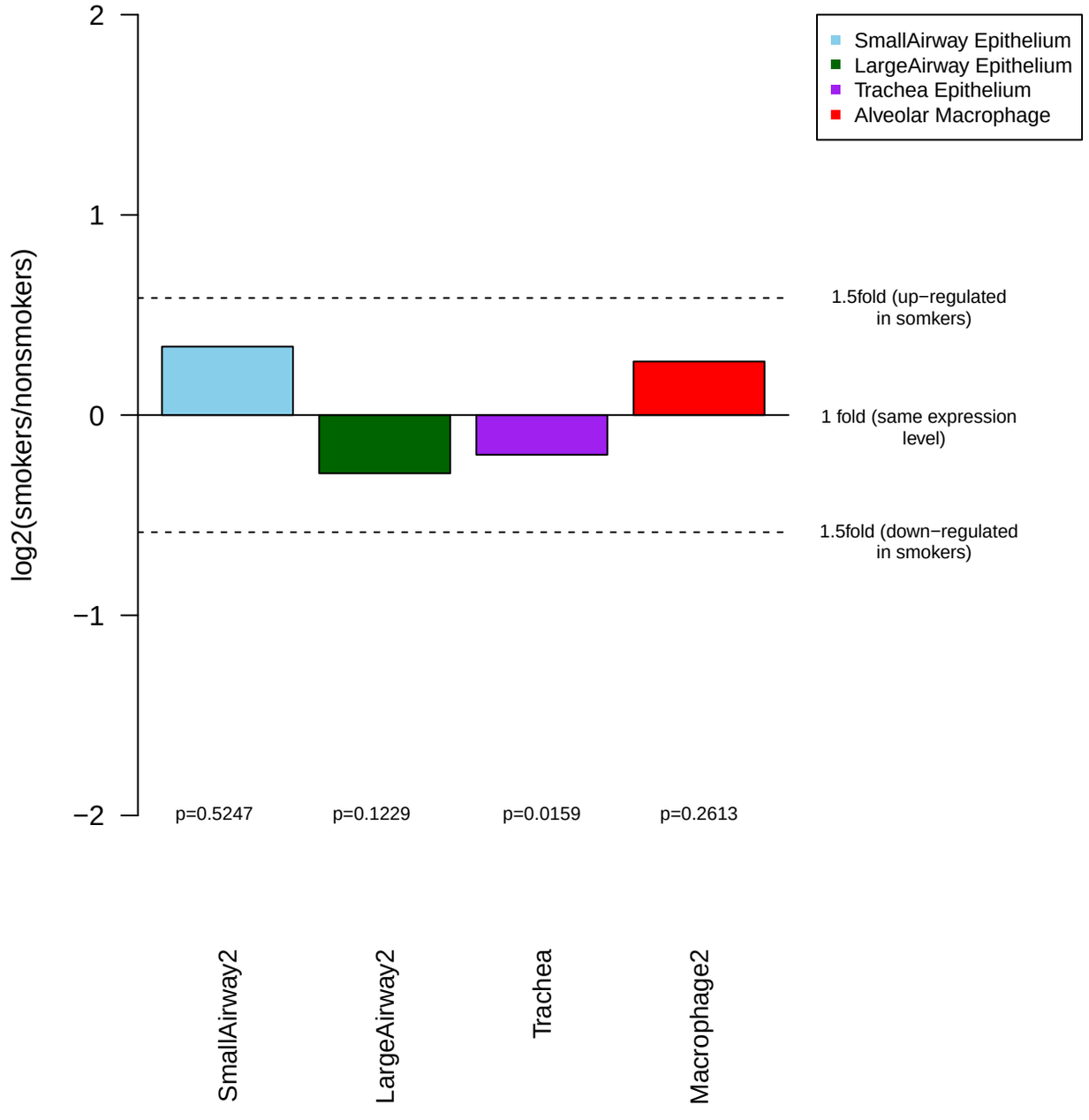
Gene expression ratio in Males

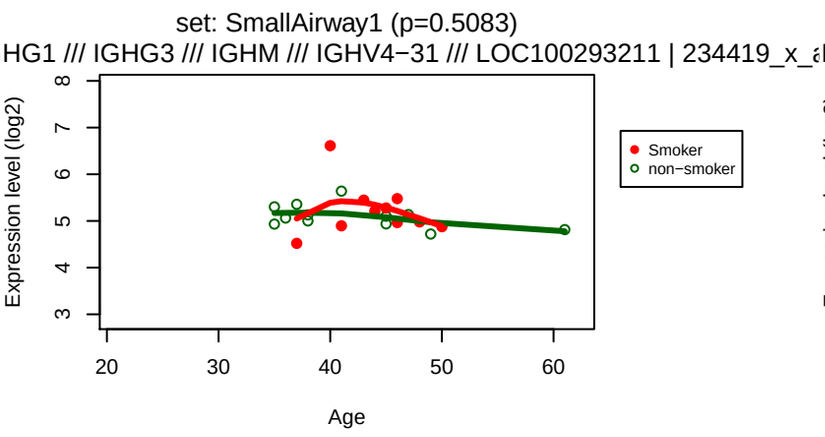
GHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_at

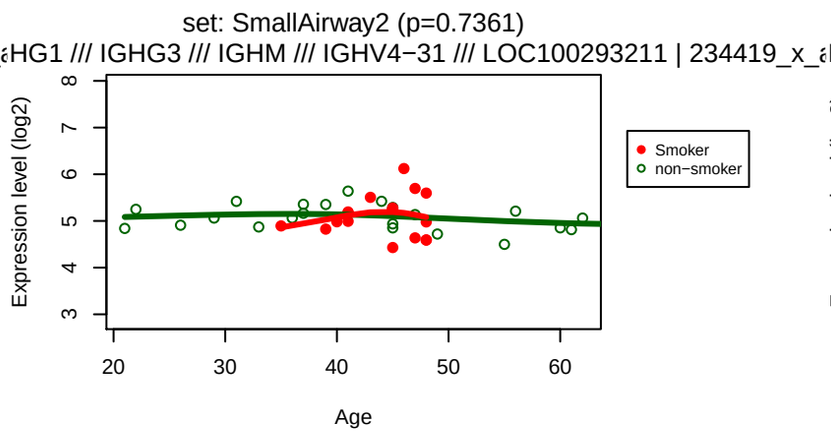


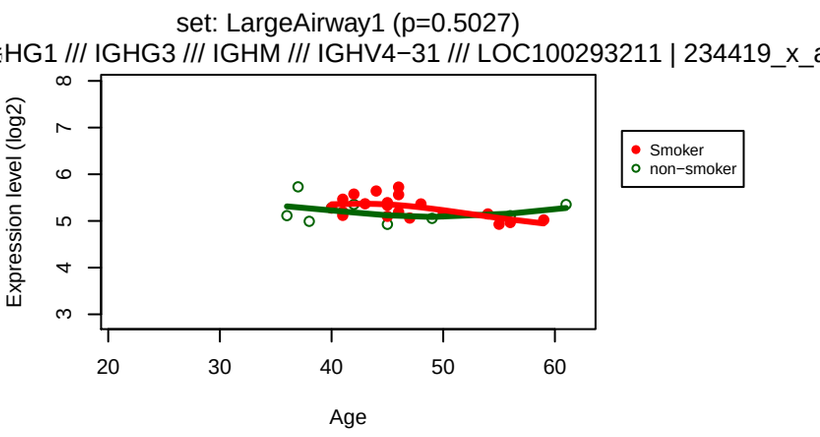
Gene expression ratio in Females

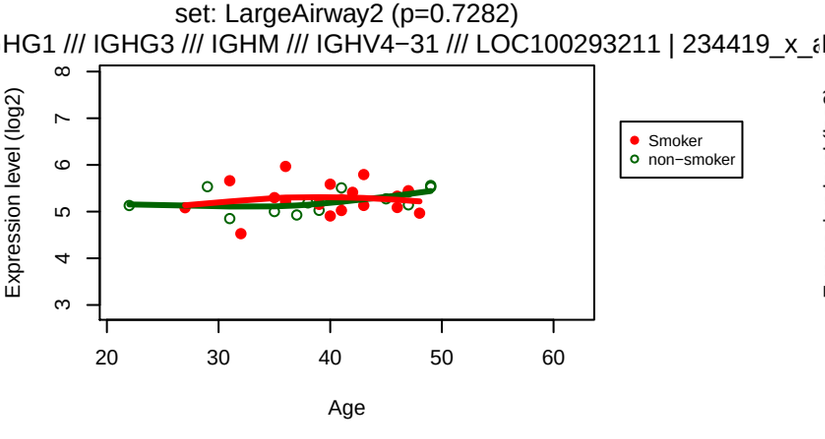
gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_at

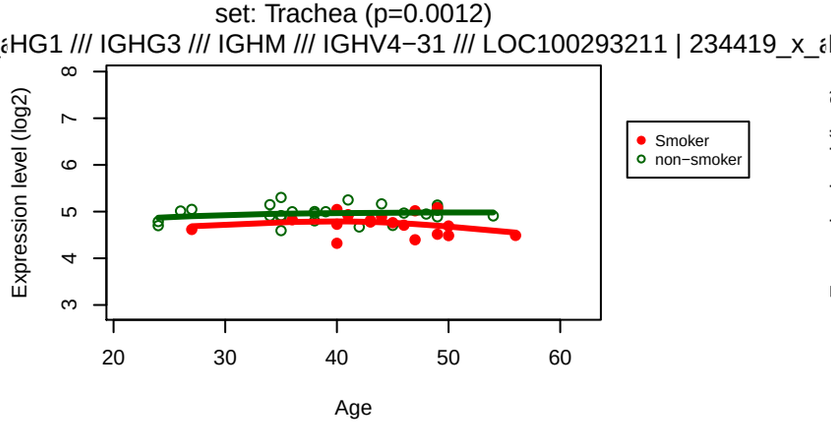


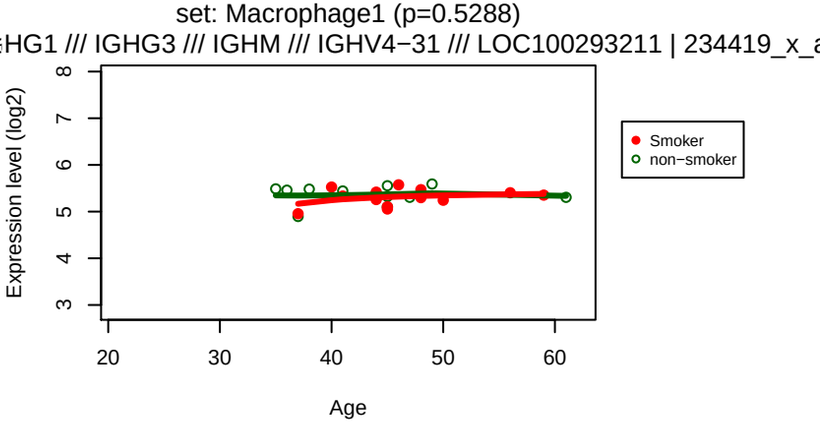

set: SmallAirway1 (p=0.5083)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_a
Expression level (log2)
Age
Smoker
non−smoker

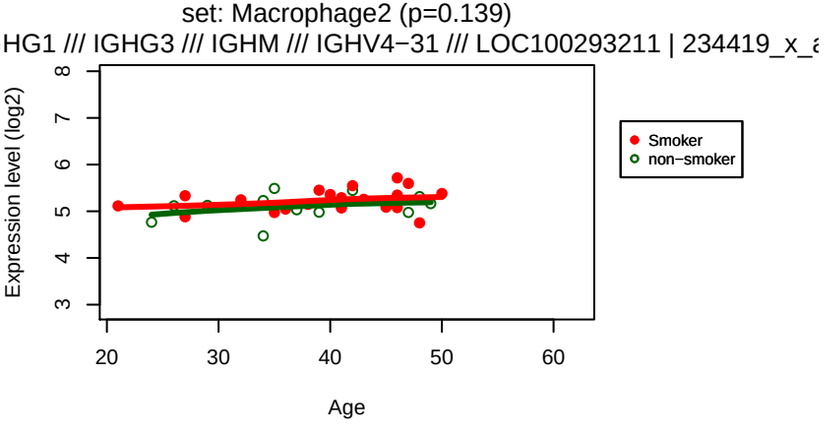

set: SmallAirway2 (p=0.7361)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_a
Expression level (log2)
Age
Smoker
non−smoker

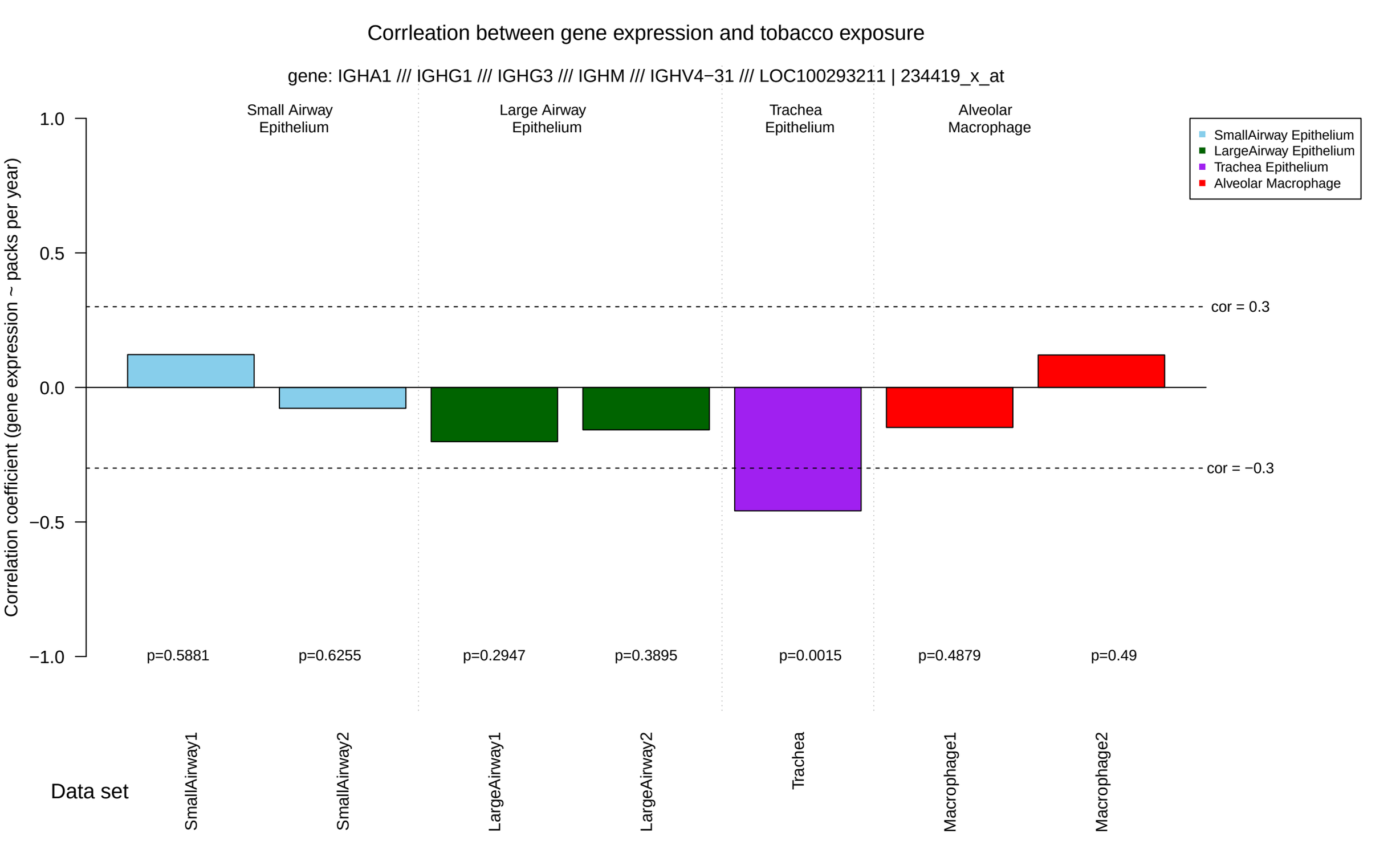

set: LargeAirway1 (p=0.5027)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_a
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7282)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_a
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_a
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5288)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_a
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.139)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_a
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100293211 | 234419_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5881
p=0.6255
p=0.2947
p=0.3895
p=0.0015
p=0.4879
p=0.49
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2